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Assignment - 3

Q1) $n=3$ $p=1/3$ and $n=5$ $p=1/3$

Obtain $P(x+y \geq 1)$

$1 - P(x+y=0) = 1 - P(x=0 \cap y=0)$

$= 1 - \left((2/3)^3 \left(2/3\right)^5\right)$

$= 1 - (2/3)^8$

Q3) (i) $X+Y=K$

$$\sum_{x=0}^{K} \frac{d_1^{x} e^{-d_1}}{x!} \times \frac{d_2^{K-x} e^{-d_2}}{(K-x)!}$$

$$= \frac{e^{-(d_1+d_2)}}{K!} \times \sum_{x=0}^{K} \frac{K!}{x!(K-x)!} \times d_1^{x} d_2^{K-x}$$

$$= \frac{e^{-(d_1+d_2)}}{K!} \times (d_1+d_2)^K = \boxed{\frac{e^{-(d_1+d_2)} (d_1+d_2)^K}{K!}}$$

(ii) $X=Y$ if $X=m$ then $Y=m$

$$P(X=Y=m) = \frac{d_1^m e^{-d_1}}{m!} \times \frac{d_2^m e^{-d_2}}{m!}$$

$$= e^{-(d_1+d_2)} \frac{(d_1 d_2)^m}{(m!)^2} = \frac{e^{-(d_1+d_2)} (d_1 d_2)^m}{(m!)^2}$$

$$P(X=Y) = \sum_{m=0}^{\infty} \frac{e^{-(d_1+d_2)} (d_1 d_2)^m}{(m!)^2}$$

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Q4) $E[e^{y}]$ where $y = \frac{x-m}{\sqrt{m}}$

$$E\left[e^{t\left(\frac{x-m}{\sqrt{m}}\right)}\right] = E\left[1 + \frac{tz}{\sqrt{m}} + \frac{t^2 z^2}{2!\,m} + \cdots\right] \quad \text{where } z = x - m$$

$$\Rightarrow 1 + E\left[\frac{tz}{\sqrt{m}}\right] + \frac{1}{2}E\left[\frac{t^2 z^2}{m}\right] + \cdots$$

$$\Rightarrow 1 + \frac{t}{\sqrt{m}}E[z] + \frac{t^2}{2m}E[z^2] + \cdots$$

$$\lim_{m\to\infty} 1 + \frac{t}{\sqrt{m}}E[x-m] + \frac{t^2}{2m}E[(x-m)^2] + \cdots$$

$$\lim_{m\to\infty} 1 + \frac{t}{\sqrt{m}} \times 0 + \frac{t^2}{2m}E[x^2 + m^2 + 2xm]$$

$$= 1 + \frac{t^2}{2m}\left(E[x^2] + m^2 - 2xm\right)$$

$$\lim_{m\to\infty} 1 + \frac{t^2}{2m}\left(m^2 + m + m^2 - 2m^2\right)$$

$$\lim_{m\to\infty} = 1 + \frac{t^2}{2} \quad \text{Ans} \quad \boxed{E[e^{y}] = 1 + \frac{t^2}{2}}$$

Q5) $\ln(x) \sim N(\mu, \sigma^2)$

$$\text{pdf}(x) = \frac{\partial F(x)}{\partial x} \Rightarrow \frac{\partial F(X \le x)}{\partial x} = \frac{\partial F(\ln x \le \ln x)}{\partial x}$$

$$= \frac{\partial}{\partial x}\Phi\left(\frac{\ln x - \mu}{\sigma}\right)$$

Φ : cdf of normal dist.
φ : pdf of normal dist.

$$= \varphi\left(\frac{\ln x - \mu}{\sigma}\right) \cdot \frac{\partial}{\partial x}\left(\frac{\ln x - \mu}{\sigma}\right)$$

normal dist

$$= \frac{1}{\sigma x}\varphi\left(\frac{\ln x - \mu}{\sigma}\right) = \boxed{\frac{1}{\sigma x}\frac{1}{2\pi}\exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)}$$

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Q5) ~~pdf~~ $pdf(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}$

$$= \int x \frac{1}{x\sigma\sqrt{2\pi}} \times e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}} dx$$

$$= \int \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right) dx$$

$$\left.\begin{aligned} &\frac{\ln x - \mu}{\sigma} = m \\ &\frac{1}{\sigma x} dx = dm \quad \Rightarrow dx = x\sigma dm \end{aligned}\right\} \text{substitution}$$

$$\int \frac{1}{\sqrt{2\pi}} e^{-m^2/2} x\sigma dm = \int \frac{1}{\sqrt{2\pi}} e^{-m^2/2} dm = \int \frac{e^{\sigma m + \mu + \frac{\sigma^2}{2}}}{\sqrt{2\pi}} dm$$

$$\Rightarrow \frac{e^{\mu}}{\sqrt{2\pi}} \int e^{\sigma m - m^2/2} dm$$

$$e^{1/2(2\sigma m - m^2)} \Rightarrow \frac{e^{-1/2(m-\sigma)^2}}{e^{-\sigma^2/2}}$$

$$\frac{e^{\mu + \sigma^2/2}}{\sqrt{2\pi}} \int e^{-1/2(m-\sigma)^2} dm = \cancel{\frac{e^{\mu+\sigma^2/2}}{\sqrt{2\pi}} \int e^{-1/2(m-\sigma)^2} dm}$$

$$\frac{e^{\mu + \frac{\sigma^2}{2}}}{\cancel{\sqrt{2\pi}}} \cdot \cancel{\sqrt{2\pi}} = e^{\mu + \sigma^2/2}$$

$$\boxed{\text{Mean} = \exp\left(\mu + \frac{\sigma^2}{2}\right)}$$

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$$E[x^2] = \int x^2 \times \frac{1}{x\sigma\sqrt{2\pi}} \times \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right] dx$$

$$= \int \frac{x}{\sigma\sqrt{2\pi}} \times \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right] dx$$

$$\therefore m = \frac{\ln x - \mu}{\sigma}$$

$$\int \frac{x^2}{\sqrt{2\pi}} \times \exp(-m^2/2) \qquad \therefore e^{m\sigma + \mu} = x$$

$$= \int \frac{\exp(2m\sigma + 2\mu - m^2/2)}{\sqrt{2\pi}}$$

$$\frac{1}{\sqrt{2\pi}} e^{2\mu} \int \exp\left(\tfrac{1}{2}(4m\sigma - m^2)\right) = \frac{e^{2\mu} e^{4\sigma^2}}{\sqrt{2\pi}} \int e^{-\frac{1}{2}(m^2 + (2\sigma)^2 - 4m\sigma)}$$

$$\Rightarrow \frac{\exp(2\mu + 2\sigma^2)}{\sqrt{2\pi}} \int e^{-\frac{1}{2}(m - 2\sigma)^2} dm = \frac{e^{2\mu + 4\sigma^2}}{\cancel{\sqrt{2\pi}}} \cancel{\sqrt{2\pi}}$$

$$= \boxed{e^{2\mu + 2\sigma^2} = E[x^2]}$$

$$Var(X) = E[x^2] - (E(X))^2$$
$$= e^{2\mu + \sigma^2} \cdot (e^{\sigma^2} - 1)$$

Q6) Let y be a r.v defined as

$$Y = \begin{cases} 0 & \text{if } x < t \\ t & x \geq t \end{cases}$$

Thus y is a discrete r.v. with pmf

$P_Y(0) = P[X < t]$ and $P_Y(t) = P(X \geq t)$

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$E[y] = 0\, p_y(0) + t\, p_y(t)$

$= t P[X \geq t]$

Note $X \geq y$ hence, $E[X] \geq E[Y]$

$E[X] \geq t P[X \geq t]$

$E[X]/t \geq P[X \geq t]$

$\therefore$ Hence, proved.

Q7) Prove $Var(ax+by+c) = a^2 Var x + b^2 Var y + 2ab\, cov(x,y)$

$\Rightarrow E[(ax+by+c)^2] - [E(ax+by+c)]^2$

$E[a^2x^2 + b^2y^2 + c^2 + 2acx + 2bcy + 2abyx] - [E(ax) + E(by) + c]^2$

assuming independence

$a^2E[x^2] + b^2E[y^2] + c^2 + 2ac\cancel{E[x]} + 2bc\cancel{E[y]} + 2ab\cancel{E[x]E[y]}$
$- a^2E^2(x) - b^2E^2[y] - c^2 - 2ab\cancel{E[x]E[y]} - 2ac\cancel{E[x]}$
$- 2bc\cancel{E[y]}$

$a^2(E[x^2] - E^2[x]) + b^2(E[y^2] - E^2[y])$

$= a^2 Var x + b^2 Var y.$

Assuming x and y are not independent.

$a^2 Var x + b^2 Var y + 2ab[E[yx] - E[x]E[y]]$

$= a^2 Var x + b^2 Var y + 2ab\, cov(X, Y)$

$\therefore$ Hence proved.

Q2) X is a Poisson variate with mean λ. Show that

$$\mu_{r+1} = r\lambda\mu_{r-1} + \lambda\frac{d\mu_r}{d\lambda}, \quad r = 1, 2, \ldots$$

where $\mu_r = E[\{x - E(x)\}^r]$

$$\mu_r = \Sigma (x-\lambda)^r = \Sigma p_i x_i$$

$$\mu_r = \Sigma (x-\lambda)^r e^{-\lambda}\frac{\lambda^x}{x!}$$

$$\frac{d\mu_r}{d\lambda} = -e^{-\lambda}\left[\Sigma (x-\lambda)^r \frac{\lambda^x}{x!}\right] + e^{-\lambda}\left[\Sigma \frac{d}{d\lambda}\left[\frac{(x-\lambda)^r \lambda^x}{x!}\right]\right]$$

$$= -e^{-\lambda}\left[\Sigma (x-\lambda)^r \frac{\lambda^x}{x!}\right] + e^{-\lambda}\left[\Sigma (x-\lambda)^r \frac{x\lambda^{x-1}}{(x!)} + \Sigma \frac{\lambda^x}{x!} r(x-\lambda)^{r-1}(-1)\right]$$

$$= -\mu_r + e^{-\lambda}\Sigma (x-\lambda)^r x \frac{\lambda^{x-1}}{x!} - r e^{-\lambda}\Sigma \frac{\lambda^x}{x!}(x-\lambda)^{r-1}$$

$$\frac{d\mu_r}{d\lambda} = -\mu_r - r\mu_{r-1} + \frac{e^{-\lambda}}{\lambda}\Sigma (x-\lambda)^r \frac{\lambda^x}{x!}$$

$$\frac{d\lambda\mu_r}{d\lambda} + r\lambda\mu_{r-1} = e^{-\lambda}\Sigma (x-\lambda)^r \frac{\lambda^x}{x!}$$

$$= e^{-\lambda}\sum_0^n (x-\lambda)^r \frac{\lambda^x}{x!}[x-\lambda] = \boxed{\frac{d\lambda\mu_r}{d\lambda} + r\lambda\mu_{r-1} = \mu_{r+1}}$$

Hence, proved.

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